Proceedings of the Second International Conference on Applied Artificial Intelligence and Computing (ICAAIC 2023)
IEEE Xplore Part Number: CFP23BC3-ART; ISBN: 978-1-6654-5630-2

Enhancing Low Resource NER using Assisting Language and Transfer Learning

Maithili Sabane *[1,3], Aparna Ranade *[1,3], Onkar Litake *[1,3], Parth Patil *[1,3],
Raviraj Joshi [2,3], Dipali Kadam [1,3]
[1]Pune Institute of Computer Technology, Pune
[2]Indian Institute of Technology Madras, Chennai
[3] L3Cube, Pune
{msabane12,aparna.ar217,litakeonkar,parthpatil8399}@gmail.com
ravirajoshi@gmail.com, ddkadam@pict.edu

2023 2nd International Conference on Applied Artificial Intelligence and Computing (ICAAIC) | 978-1-6654-5630-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICAAIC56838.2023.10141204

***Abstract*—Named Entity Recognition (NER) is a fundamental task in NLP that is used to locate the key information in text and is primarily applied in conversational and search systems. In commercial applications, NER or comparable slot filling methods have been widely deployed for popular languages. NER is utilized in applications such as human assets, client benefit, substance classification, and the scholarly community. This research study focuses on identifying name entities for low-resource Indian languages that are closely related, like Hindi and Marathi. This study uses various adaptations of BERT such as baseBERT, AlBERT, and RoBERTa to train a supervised NER model. The, compares multilingual models with monolingual models and establish a baseline. The results show the assisting capabilities of the Hindi and Marathi languages for the NER task. Also, the results show that the models trained using multiple languages perform better than a single language. However, this research study also observe that blind mixing of all datasets doesn't necessarily provide improvements and data selection methods may be required.**

***Keywords*—NER, transformers, low resource, transfer learning**

I. Introduction

A prominent method in natural language processing(NLP) for information extraction is NER(Named Entity Recognition) [1]. It was introduced in 1995. This technique comprises detecting a named entity and then classifying it. These entities include names of places, people, and organisations, as well as numerical quantities such as money, time, and dates.

The process of Named Entity Recognition (NER) is performed using three major techniques [2]. These are rule-based NER, machine-learning NER, and hybrid NER. Rule-based NER is a technique in which linguists formalise rules. It includes a gazetteer, lexicalized grammar, etc. Machine-learning-based NER is a method in which models are trained using annotated text. Hidden Markov Models (HMMs), Conditional Random Fields (CRFs), and other machine-learning-based approaches are examples. Deep learning architectures have since been investigated for NER which includes architectures such as Convolutional Neural Network (CNN) [3], Long Short Term Memory (LSTM) [4], and Bi-Directional Long Short Term Memory (Bi-LSTM) [5]. Hybrid NER uses a combination of both of these methodologies.

Previously, much work on NER in the English language was done, resulting in a vast number of resources for the English language. Indian languages have a small amount of NER literature. This is mainly because of data sparsity and lack of tools owing to the intricacies of the languages. These intricacies make it difficult for the existing algorithms to be applied to low-resource languages.

To overcome these difficulties for low-resource languages and compensate for the lack of data, cross-lingual transfer training has been utilized in the domain of named entity recognition. This strategy focuses on retaining and transferring information obtained while solving one problem to another that is separate yet related. It entails developing a model on huge corpora in one language and exchanging that knowledge with a model built on a limited amount of data in another. This makes it easier to enhance the entity classification of low-resource languages by taking advantage of the data from different languages. It is commonly known as low-resource cross-lingual transfer learning [6]. The motivation was as very low resource and data is available for low resource indian

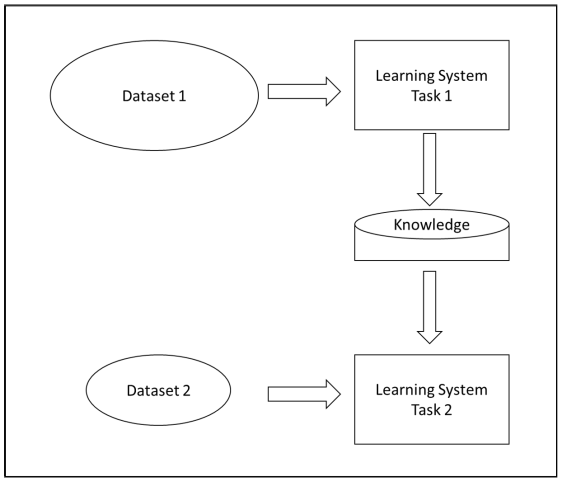

Fig. 1. Model Architecture

* Equal contribution of the authors.

Authorized licensed use limited to: Ramarao Adik Institute of Technology. Downloaded on February 09,2026 at 11:02:37 UTC from IEEE Xplore. Restrictions apply.

languages. We are establishing a benchmark for a low resource indian language which has significantly less amount of data and assist it with transfer learning with similar low resource language. This can assist improve the model's performance. We have compared the performance of multilingual BERT models trained on Marathi and Hindi datasets to that of their monolingual counterparts. These two languages follow the same script (Devanagari) and have similar grammatical rules and significant overlapping words. This motivated us to perform the cross-lingual evaluation using these languages and benchmark their assisting capabilities. This work's main contribution is as follows:

- We offer a detailed study of the architectures for transfer learning NER.
- We compare the outcomes of deep learning models trained using typical monolingual versus multilingual techniques.
- We demonstrate the benefits of cross-lingual transfer learning and how it may help improve scores in Hindi and Marathi.

The rest of the paper is organised as follows. Section 2 investigates developments in Named Entity Recognition, with a focus on Indian languages. How we set up our tests to investigate various models and experiments have been defined in section 3. Section 4 summarises the findings of all of the tests. Section 5 summarises our paper's conclusion.

II. Related Work

[7] Named Entity Recognition was first proposed at Message Understanding Conferences (MUC) in the United States in 1995. The study was widespread for Indian languages around 2008. A study was carried out using various traditional machine learning algorithms and deep learning techniques.A comparison of Conditional Random Field (CRF) and Support Vector Machine (SVM) trained on the same data was conducted by [8]. The CRF model was shown to be better. When doing NER, hybrid systems comprising of the Hidden Markov Model, MaxEnt, and handcrafted rules were shown to be more accurate as per [9]. As technology advanced, deep learning models were used to solve the NER problem. [3]Convolutional Neural Network (CNN)(d), [10], [11] and [12] were among the most popular models used for NER.

In his paper, [13] describes various NER techniques used for Indian languages so far. The research analyses and contrasts methodologies for recognising named items. It contrasts Conditional Random Field (CRF) and the Hidden Markov Model (HMM) approaches for Indian languages, concluding that the CRF method is the most successful. [14] worked on NER for the Hindi language. Machine Learning (ML), Rule-based, and Hybrid techniques are examined. Current NER algorithms have been taught to work on specific datasets and do not yield results on large datasets. As a result, the study attempts to identify flaws in existing NER systems, notably in the Hindi language. According to the article, while the machine learning technique is more systematic when predicting unknown things, it has lower accuracy than the rule-based system. Obstacles and concerns regarding NER for Marathi were discussed by [15] along with its importance. It also looks at various tactics and methodologies for creating the learning resources needed for extracting NE(Named Entities) from unstructured natural language input.

[16] demonstrates how variations in tag distributions of regularly occurring named things across primary and supporting languages impact the efficacy of multilingual learning. To address this issue, the study proposes a metric based on symmetric KL divergence that use neural networks like CNNs and Bi-LSTMs to filter out extremely contrasting training samples in the assisting language.

[17] generated baseline figures for a variety of publicly available Hindi and Marathi datasets. [15] provides a publicly available golden dataset including 25000 phrases in Marathi. In [18] research, they overcome the problem of data sparsity by stealing attributes from data from a closely related language. A supervised NER system is trained using hierarchical neural networks. The network's common layers are used to borrow a feature from a closely comparable language. When a neural network is trained on a combined dataset comprising a low-resource language and a closely related language, this is referred to as multilingual learning. Our work is similar to theirs as we combine various datasets for Hindi and Marathi languages and compare them against mono-lingual models.

III. Experimental Setup

TABLE I
IIT Bombay corpora's individual tags count

Tags	Train	Test	Validation
O	61235	28215	7349
NEL	4821	3148	796
NEP	1546	760	194
NEO	173	91	31

TABLE II
IJCNLP individual tags count

Tags	Train	Test	Validation
O	199481	42655	42068
NEL	3257	692	880
NEP	3893	886	806
NEO	2119	459	392

TABLE III
Wiki ANN Marathi individual tag count

Tags	Train	Test
O	46011	19633
NEL	8711	3958
NEP	1115	4671
NEO	10129	4310

We are confining our Named Entity Recognition (NER) efforts to Marathi and Hindi, two of the top three languages spoken in India, and performed Named Entity Recognition on all publicly available datasets.

Authorized licensed use limited to: Ramarao Adik Institute of Technology. Downloaded on February 09,2026 at 11:02:37 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
WIKI ANN HINDI INDIVIDUAL TAGS COUNT

Tags	Train	Test
O	20015	9243
NEL	4469	2768
NEP	13015	2678
NEO	11102	6150

We are using a dataset released in IJCNLP in 2008 as well as the WikiAnn NER Corpus published by [19] in 2017 for the Hindi language. The IJCNLP dataset has 11,400 sentences in total. There are 12 categories in the IJCNLP dataset: organization, person, title-object, number, abbreviation, location, brand, title-person, time, measurement, designation, and words. As no data split was provided, we divided the data into 70-15-15 train, test, and tune segments. [19] dataset for Hindi which is a silver standard dataset consisting of three categories: Location, Person, and Organization containing a total of 11,833 sentences. We are utilizing the IIT Bombay Marathi NER dataset provided by [16] in 2018 and the WikiAnn NER Corpus dataset given by [19] in 2017 for the Marathi language. There are a total of 5,591 sentences in the IIT B dataset. It is divided into three categories: location, person, and organization. The dataset is already separated into train-test-tune segments. [19] dataset for marathi which is a silver standard dataset contains 14,978 phrases divided into three categories: Location,Organisation and Person. Both datasets are in the IOB format.

The difficulties encountered while dealing with these datasets were as follows. Both datasets consisted of English terms. Tagging in the IJCNLP dataset was non-uniform, and the dataset had some perplexing tags. Marathi NER Corpus IIT Bombay contains, more than 39 percent of sentences contained O tags, and in IJCNLP Corpus, more than 68 percent of the sentences contained O tags. Several Marathi and Hindi datasets are available for NER, however, they are not publicly accessible.

For uniformity and consitency of training the models we have considered only location(NEL), person(NEP), and organisation(NEO) as tags. For the IJCNLP dataset we have considered only 'NEP' 'NEO' 'NEL' as tags and remaining tags 'NETI', 'NETE', 'NEA', 'NED', 'NEM', 'NEN', 'NETO' were replaced with 'O'. For the IIT Bombay NER dataset, we have removed the IOB notation and converted them into non-IOB format for example, 'B-PERSON' 'I-PERSON' has been converted to 'NEP' 'NEP' and 'B-ORGANISATION' 'I-ORGANISATION' to 'NEO' 'NEO' and 'B-LOCATION' 'I-LOCATION' to 'NEL' 'NEL'. Similarly, for the wiki-ann Marathi dataset 'B-ORG' 'I-ORG' to 'NEO' 'NEO' and 'B-PER' 'I-PER' to 'NEP' 'NEP' and 'B-LOC' 'I-LOC' to 'NEL' 'NEL'. For Wiki-ann Hindi dataset, 'B-ORG' 'I-ORG' to 'NEO' 'NEO' and 'B-PER' 'I-PER' to 'NEP' 'NEP' and 'B-LOC' 'I-LOC' to 'NEL' 'NEL'. We have merged the IJCNLP train set and the IIT Bombay Marathi train set, the IJCNLP tune set, and IIT Bombay Marathi tune set and then reshuffled the sentences and evaluated the whole on various transformer-based models like mBert, Indicbert, Xlmroberta, etc by testing it individually on the IIT Bombay test set and IJCNLP test set. Similarly, for the Wiki-Ann corpus, we have merged the Wiki-Ann Hindi train set and Wiki-Ann Marathi train set and evaluated models by testing individually on the Wiki-Ann Marathi test set Wiki-Ann Hindi test set. In the end, we merged all four datasets i.e, IJCNLP, IIT Bombay Marathi dataset, Wiki-Ann Marathi, and Wiki-Ann Hindi, and evaluated the results individually on each test set.

IV. EXPERIMENTAL TECHNIQUES

A. Model Architecture

A type of deep learning model Transformers are models which can process sequential data, such as natural language, by learning representations of the data that capture its underlying structure. In a transformer-based NER model, the transformer architecture is used to encode the input text into a sequence of hidden representations, which are then fed into a classifier to predict the named entities. The context is detected by the transformer's ability to capture the relationships between the words in the input text, which is learned through a self-attention mechanism.

Self-attention allows the transformers which can learn to recognize the context such as location, organisation or name of person in which named entities appear.

Once the transformer has learned to recognize the relevant context for named entities, the classifier can use this information to make more accurate predictions. Typically, this technique is trained on a huge dataset of labelled samples, where the model learns to recognise the context of named things in input text and correlate them with the appropriate label. An "attention" mechanism in transformers helps to examine a connection between all of the words in a sentence. This aids in establishing the process of word evaluation to show which phrase components are most important to offer differential weightings and thus explains why ambiguous elements are solved quickly and efficiently. The context is recognised by transformer where it recognises the meaning of each word in the statement. The training time decreases as the feature improves parallelization.

BERT: BERT [20] is a transformer-based approach created by Google for NLP pre-training referred to as Bidirectional Encoder Representations from Transformers. Using unlabeled text by focusing on both sides of context left and right simultaneously across all layers, BERT pre-trains deep bidirectional representations. BERT models are fine-tuned with just one more output layer to offer high-performing models for a wide range of jobs.

mBERT: mBERT [21], referred to as multilingual BERT has been trained on and can be used in 104 languages. When mBERT was built, data from all 104 languages were combined. As a result, at the same time, mBERT incorporates and comprehends word connections in all 104 languages.

ALBERT: A Lite Bert (ALBERT) [22] is a MLP model uses less parameters while retaining accuracy which is de-

Authorized licensed use limited to: Ramarao Adik Institute of Technology. Downloaded on February 09,2026 at 11:02:37 UTC from IEEE Xplore. Restrictions apply.

TABLE V
NUMBER OF SENTENCES AND TAGS COUNT IN THE DATASETS

Dataset Name	Sentences count			Tags count		
	Train	Test	Tune	Train	Test	Tune
IJCNLP 2008 NER Corpus	7979	1711	1710	208750	44692	44146
Marathi NER IIT Bombay Corpus	3588	1533	470	67775	32214	8370
WikiAnn NER (Marathi) Corpus	10674	4304	–	76006	32572	–
WikiAnn NER (Hindi) Corpus	8356	3477	–	48601	20829	–

TABLE VI
F1 SCORES FOR THE MODELS TESTED ON MERGED HINDI & MARATHI DATASETS AND INDIVIDUAL DATASET. IJCNLP(H) = IJCNLP HINDI DATASET, IITB(M) = IITB MARATHI DATASET, WIKI(H) = WIKIANN HINDI, WIKI(M) = WIKIANN MARATHI

Test Datasets	**Train Datasets**	**Mbert**	**Indic Bert**	**Xlm Roberta**	**Maha Albert**	**Roberta Hindi**	**Maha Bert**	**Maha Roberta**
IITB Marathi	IJCNLP(H) &IITB(M)	60.9	63.12	**64.36**	63.05	**45.05**	**65.29**	**64.54**
	IITB(M)	61.45	63.26	60.97	63.43	43.27	62.07	64.18
IJCNLP Hindi	IJCNLP(H) &IITB(M)	72.7	**76.85**	**80.34**	**75.61**	**70.02**	**76.9**	**81.3**
	IJCNLP(H)	75.52	76.32	79.71	63.11	70.01	76.53	78.67
WikiANN Marathi	Wiki(H&M)	**87.3**	86.86	**87.47**	86.33	**83.52**	**88.58**	87.94
	Wiki(M)	86.49	87.03	87.38 7	87.15	82.5	88.18	88.9
WikiANN Hindi	Wiki(H&M)	**83.85**	**84.26**	**85.94**	**83.33**	**82.88**	**82.83**	**85.9**
	Wiki(H)	81.21	82.65	83.04	81.68	80.52	81.95	80.66

TABLE VII
F1 SCORES FOR THE MODELS TESTED BY MERGING ALL DATASETS AND INDIVIDUAL DATASET

Model Name	**Dataset type**	**IIT Bombay**	**IJCNLP Hindi**	**WikiANN Marathi**	**WikiANN Hindi**
mbert	Merged	57.59	74.83	85.48	**83.85**
	Mono	61.45	75.52	86.49	81.21
Indic Bert	Merged	59.62	74.64	86.33	**83.59**
	Mono	63.26	76.32	87.03	82.65
Xlm Roberta	Merged	60.73	78.58	86.78	84.21
	Mono	60.97	79.71	87.38	83.04
MahaBert	Merged	**62.76**	74.98	86.98	**83.19**
	Mono	62.07	76.53	88.18	81.95
MahaRoberta	Merged	64.11	**79.22**	87.17	**84.36**
	Mono	64.18	78.67	88.9	80.66
Maha Albert	Merged	58.12	**72.38**	85.64	79.86
	Mono	63.43	63.11	87.15	81.68
Roberta Hindi	Merged	**44.65**	68.72	**83.01**	**82.11**
	Mono	43.27	70.01	82.5	80.52

veloped by google. ALBERT offer higher data throughput and can train about 1.7 times faster than BERT models. On a large-scale corpus of 12 main Indian languages multilingual ALBERT model is trained which is referred to as IndicBERT. IndicBERT contains many fewer parameters yet it achieves comparable or superior performance to other models.

RoBERTa: RoBERTa [23] is a self-supervised model and huge english data. It was trained primarily with the Masked language modeling (MLM) objective. The MLM goal is to mask out certain tokens in the input text and train the model to anticipate the missing tokens. RoBERTa improves on the original BERT model's architecture by including more training data, omitting the next sentence prediction job, and fine-tuning the hyperparameters.

V. RESULTS

We now discuss our results for the multilingual models trained on both Hindi and Marathi NER datasets. The aim is to evaluate assisting capability of these languages in improving individual language performance.

Initially, we merged the IIT Bombay(Marathi) with IJCNLP(Hindi) for training and tested it individually with the IIT Bombay and IJCNLP datasets. We then compared it with the results obtained on models trained with single monolingual

Authorized licensed use limited to: Ramarao Adik Institute of Technology. Downloaded on February 09,2026 at 11:02:37 UTC from IEEE Xplore. Restrictions apply.

data. The observations portray that XLM Roberta, RoBERTa Hindi, MahaBERT, and Maha Roberta perform better on the mixed dataset than on the monolingual IIT Bombay dataset (Marathi). Similarly, when the mixed data model is tested on IJCNLP (Hindi), the scores observed are better for all the models except with mBERT where monolingual IJCNLP performs better.

Likewise, Wiki ANN datasets for Hindi and Marathi were merged and tested with individual datasets to compare the results with models that were trained and tested on monolingual data. Again the majority of the models, when tested with individual datasets, outperformed the models trained on monolingual datasets of Wiki ANN.

Lastly, we merged all the datasets, viz. IIT Bombay(Marathi), IJCNLP(Hindi), and Wiki ANN(Hindi and Marathi) and tested them individually on the merged dataset. We see consistent improvements in Wiki ANN(Hindi) dataset using this expanded training set. However, the improvements are not consistently visible for other datasets. This shows that blind mixing of all the available datasets may not be desirable, and some data selection methods may be further employed. The differene in the domain of Wiki ANN dataset and other datasets is possibly responsible for this behavour. The difference in the domain of the Wiki ANN dataset and other datasets is possibly responsible for this behavior and needs further investigation. NER Models performance can be done by increasing the number and quality of training data. To understand patterns and relationships between words and named items, NER models rely significantly on training data and training data quality have been demonstrated to improve NER performance which can help capture linguistic and contextual nuances, which can help improve NER performance.Pre-trained language models can be improved by fine-tuning them on specific NER tasks. Incorporating external knowledge sources, such as gazetteers, dictionaries, and ontologies, can help to improve NER performance by providing additional information on named entities that may not be included in the training data. Ensemble learning can be accomplished by training numerous models on distinct subsets of the training data, or by mixing models of various sorts, such as rule-based models, statistical models, and neural network models.

VI. Conclusion

The primary purpose of this effort is to create multilingual deep learning-based models for named entity identification in Indian languages. We aim to benchmark multilingual transformer-based models on Hindi and Marathi NER datasets. We also explore the assisting capabilities of these languages. We show that combining the two languages does provide improvements in individual languages. However, mixing all the available datasets results in slight degradation of the performance on most of the individual datasets.

References

[1] P. Sun, X. Yang, X. Zhao, and Z. Wang, "An overview of named entity recognition," in *2018 International Conference on Asian Language Processing (IALP)*, 2018, pp. 273–278.

[2] J. Li, A. Sun, J. Han, and C. Li, "A survey on deep learning for named entity recognition," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 1, pp. 50–70, 2022.

[3] S. Albawi, T. A. Mohammed, and S. Al-Zawi, "Understanding of a convolutional neural network," in *2017 International Conference on Engineering and Technology (ICET)*, 2017, pp. 1–6.

[4] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, pp. 1735–80, 12 1997.

[5] R. Panchendrarajan and A. Amaresan, "Bidirectional lstm-crf for named entity recognition," 05 2019.

[6] S. Schuster, S. Gupta, R. Shah, and M. Lewis, "Cross-lingual transfer learning for multilingual task oriented dialog," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 3795–3805. [Online]. Available: https://aclanthology.org/N19-1380

[7] R. Grishman and B. Sundheim, "Message Understanding Conference-6: A brief history," in *COLING 1996 Volume 1: The 16th International Conference on Computational Linguistics*, 1996. [Online]. Available: https://aclanthology.org/C96-1079

[8] A. A. Krishnarao, H. Gahlot, A. Srinet, and D. S. Kushwaha, "A comparative study of named entity recognition for hindi using sequential learning algorithms," *2009 IEEE International Advance Computing Conference*, pp. 1164–1169, 2009.

[9] R. Srihari, "A hybrid approach for named entity and sub-type tagging," in *Sixth Applied Natural Language Processing Conference*. Seattle, Washington, USA: Association for Computational Linguistics, Apr. 2000, pp. 247–254. [Online]. Available: https://aclanthology.org/A00-1034

[10] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 11 1997. [Online]. Available: https://doi.org/10.1162/neco.1997.9.8.1735

[11] G. Yang and H. Xu, "A residual bilstm model for named entity recognition," *IEEE Access*, vol. 8, pp. 227 710–227 718, 2020.

[12] M. Arkhipov, M. Trofimova, Y. Kuratov, and A. Sorokin, "Tuning multilingual transformers for language-specific named entity recognition," in *Proceedings of the 7th Workshop on Balto-Slavic Natural Language Processing*. Florence, Italy: Association for Computational Linguistics, Aug. 2019, pp. 89–93. [Online]. Available: https://aclanthology.org/W19-3712

[13] P. Rana, S. Gupta, and K. Dutta, "Named entity recognition (ner) for hindi," *International Journal of Computer Sciences and Engineering*, vol. 6, pp. 856–859, 07 2018.

[14] K. Bhattacharjee, S. S, S. Mehta, A. Kumar, R. V. K. Mehta, D. Pandya, P. Chaudhari, and D. Verma, "Named entity recognition: A survey for indian languages," *2019 2nd International Conference on Intelligent Computing, Instrumentation and Control Technologies (ICICICT)*, vol. 1, pp. 217–220, 2019.

[15] P. Patil, A. Ranade, M. Sabane, O. Litake, and R. Joshi, "L3cube-mahaner: A marathi named entity recognition dataset and bert models," 04 2022.

[16] R. Murthy, A. Kunchukuttan, and P. Bhattacharyya, "Judicious selection of training data in assisting language for multilingual neural NER," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Melbourne, Australia: Association for Computational Linguistics, Jul. 2018, pp. 401–406. [Online]. Available: https://aclanthology.org/P18-2064

[17] O. Litake, M. Sabane, P. Patil, A. Ranade, and R. Joshi, "Mono vs multilingual bert: A case study in hindi and marathi named entity recognition," 03 2022.

[18] R. Murthy, M. M. Khapra, and P. Bhattacharyya, "Improving ner tagging performance in low-resource languages via multilingual learning," *ACM Trans. Asian Low-Resour. Lang. Inf. Process.*, vol. 18, no. 2, dec 2018. [Online]. Available: https://doi.org/10.1145/3238797

[19] X. Pan, B. Zhang, J. May, J. Nothman, K. Knight, and H. Ji, "Cross-lingual name tagging and linking for 282 languages," in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Vancouver, Canada: Association

Authorized licensed use limited to: Ramarao Adik Institute of Technology. Downloaded on February 09,2026 at 11:02:37 UTC from IEEE Xplore. Restrictions apply.

for Computational Linguistics, Jul. 2017, pp. 1946–1958. [Online]. Available: https://aclanthology.org/P17-1178

[20] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," 2019.

[21] T. Pires, E. Schlinger, and D. Garrette, "How multilingual is multilingual bert?" 2019.

[22] Z. Lan, M. Chen, S. Goodman, K. Gimpel, P. Sharma, and R. Soricut, "Albert: A lite bert for self-supervised learning of language representations," 2020.

[23] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "Roberta: A robustly optimized bert pretraining approach," 2019.

Authorized licensed use limited to: Ramarao Adik Institute of Technology. Downloaded on February 09,2026 at 11:02:37 UTC from IEEE Xplore. Restrictions apply.